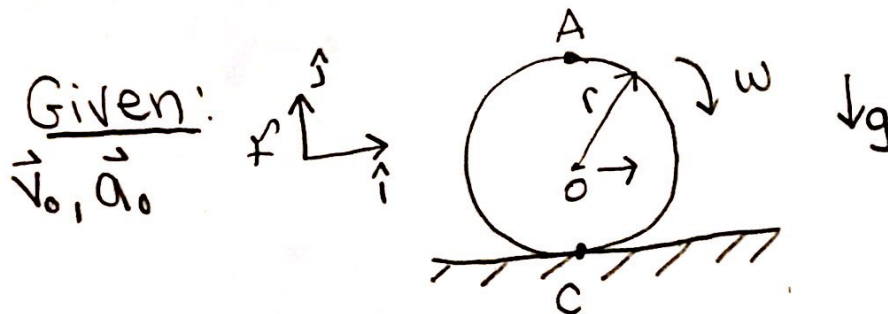


Q-Exam problem: Question 2 pg 3



Assumptions:

- the object is rolling without slipping (ie $\vec{v}_0 = -\omega r \hat{i}$)
- $\vec{v}_0, \vec{a}_0$ are unknown, can be a function of t

Find: $\vec{v}_c, \vec{a}_c, \vec{\omega}, \vec{v}_a, \vec{a}_a$

Soln:

Use kinematics and relative velocity:

$$\vec{v}_c = \vec{v}_{c/o} + \vec{v}_0$$

If roll without slip: $\vec{v}_0 = -\omega r \hat{i}$

transport theorem: $\vec{v}_{c/o} = \frac{d}{dt} \vec{r}_{c/o} = \omega \hat{k} \times \vec{r}_{c/o} = \omega \hat{k} \times -r \hat{j} = r\omega \hat{i}$

$$\therefore \boxed{\vec{v}_c = \vec{v}_0 + r\omega \hat{i} = -\omega r \hat{i} + r\omega \hat{i} = 0} \leftarrow \text{roll w/o slip}$$

$$\vec{a}_c = \vec{a}_{c/o} + \vec{a}_0$$

$$\vec{a}_{c/o} = \frac{d}{dt} \vec{v}_{c/o} = \frac{d}{dt} (r\omega \hat{i}) = r\dot{\omega} \hat{i} + r\omega^2 \hat{j}$$

$$\vec{a}_0 = \frac{d}{dt} \vec{v}_0 = \frac{d}{dt} (-\omega r \hat{i}) = -\dot{\omega} r \hat{i} - \omega^2 r \hat{j}$$

$$\therefore \boxed{\vec{a}_c = r\dot{\omega} \hat{i} + r\omega^2 \hat{j} - r\dot{\omega} \hat{i} - r\omega^2 \hat{j} = 0}$$

$$\vec{V}_A = \vec{V}_{A/O} + \vec{V}_O$$

$$\vec{V}_A = \omega \hat{k} \times \vec{r}_{A/O} = \omega \hat{k} \times r \hat{j} = -r\omega \hat{i}$$

$$\vec{V}_O = -\omega r \hat{i}$$

$$\therefore \vec{V}_A = -\omega r \hat{i} - \omega r \hat{i} = -2\omega r \hat{i}$$

$$\vec{a}_A = \vec{a}_O + \vec{a}_{A/O}$$

$$\vec{a}_A = \frac{d}{dt} \vec{V}_A = \frac{d}{dt} (-2\omega r \hat{i}) = \boxed{-2r\dot{\omega} \hat{i} - 2r\omega^2 \hat{j}}$$